

# Uhlenbeck Integrals, a Rational Lax Matrix, and Compact Liouville Fibers for the Neumann Sphere

Route-A source-local certificate HCS-C349

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## Abstract

For the anisotropic Neumann oscillator on the round two-sphere we prove global completeness and derive all three Uhlenbeck integrals with their two linear relations. A rational  $2 \times 2$  Lax matrix generates the integrals as determinant residues. Direct Dirac-bracket involution and compactness close the regular Liouville two-tori and their necessary-and-sufficient rational-frequency return criterion. The six axial equilibria, invariant coordinate circles, repeated-spectrum  $SO(2)$  faces, isotropic great circles, and natural compact quantization are closed separately. Exact finite receipts test conventions but do not replace the continuum proof.

## 1 Frozen constrained system and theorem

Let  $A = \text{diag}(a_1, a_2, a_3)$  with  $a_1 < a_2 < a_3$ . On

$$T^*S^2 = \{(x, p) \in \mathbb{R}^3 \times \mathbb{R}^3 : |x|^2 = 1, x \cdot p = 0\}$$

take

$$H = \frac{1}{2}(|p|^2 + x^T A x), \quad \dot{x} = p, \quad \dot{p} = -A x + \alpha x, \quad \alpha = x^T A x - |p|^2. \quad (1)$$

Put  $L_{ij} = x_i p_j - x_j p_i$  and

$$F_i = x_i^2 + \sum_{j \neq i} \frac{L_{ij}^2}{a_i - a_j}. \quad (2)$$

**Theorem 1.** *The flow (1) is complete. The  $F_i$  are conserved, pairwise Dirac–Poisson commuting, and satisfy*

$$\sum_i F_i = 1, \quad \sum_i a_i F_i = 2H.$$

*They are the residues of the rational Lax determinant in Section 3. Every connected regular common fiber of two independent integrals is a Liouville two-torus; the physical flow on it is periodic precisely when its frequency vector has a common period. The axial, coordinate, repeated-spectrum, and isotropic clauses in Section 4 exhaust the declared parameter boundaries. The source Hamiltonian has the natural self-adjoint compact-resolvent quantization stated there.*

## 2 Completeness and the Uhlenbeck identities

The two constraint derivatives are

$$\frac{d}{dt}|x|^2 = 2x \cdot p, \quad \frac{d}{dt}(x \cdot p) = |p|^2 - x^T A x + \alpha = 0$$

on  $T^*S^2$ . Also  $\dot{H} = 0$ . On  $H = E$  the identity  $|p|^2 = 2E - x^T A x$  bounds  $p$ ; the orbit remains in a compact subset, and the smooth vector field is complete.

The angular momenta satisfy

$$\dot{L}_{ij} = (a_i - a_j)x_i x_j. \quad (3)$$

Hence

$$\dot{F}_i = 2x_i \left( p_i + \sum_{j \neq i} L_{ij} x_j \right) = 0,$$

because the parenthesis equals  $p_i + x_i(x \cdot p) - p_i|x|^2$ . Pairing  $(i, j)$  and  $(j, i)$  proves  $\sum F_i = 1$ . The weighted pairing and the Gram identity give

$$\sum_i a_i F_i = x^T A x + \sum_{i < j} L_{ij}^2 = x^T A x + |p|^2 = 2H.$$

This is the *compact Uhlenbeck owner* added in revision round zero.

### 3 Resolvent Lax generator

For  $\lambda \notin \{a_1, a_2, a_3\}$  let

$$R_\lambda = (\lambda I - A)^{-1}, \quad U = x^T R_\lambda x, \quad V = x^T R_\lambda p, \quad W = 1 + p^T R_\lambda p$$

and define

$$\mathcal{L} = \begin{pmatrix} V & U \\ -W & -V \end{pmatrix}, \quad \mathcal{M} = \begin{pmatrix} 0 & 1 \\ \alpha - \lambda & 0 \end{pmatrix}. \quad (4)$$

Using  $a_i/(\lambda - a_i) = \lambda/(\lambda - a_i) - 1$  in (1) gives

$$\dot{U} = 2V, \quad \dot{V} = W + (\alpha - \lambda)U, \quad \dot{W} = 2(\alpha - \lambda)V.$$

These are exactly  $\dot{\mathcal{L}} = [\mathcal{L}, \mathcal{M}]$ . Moreover,

$$\det \mathcal{L} = UW - V^2 = \sum_{i=1}^3 \frac{F_i}{\lambda - a_i}. \quad (5)$$

Indeed, the residue at  $a_i$  is (2); both rational functions vanish at infinity and have no other poles.

The constrained bracket is

$$\{x_i, x_j\}_D = 0, \quad \{x_i, p_j\}_D = \delta_{ij} - x_i x_j, \quad \{p_i, p_j\}_D = x_j p_i - x_i p_j.$$

Differentiating (2), substituting these three coordinate brackets, and pairing the denominators  $a_i - a_j$  and  $a_j - a_i$  gives

$$\{F_i, F_j\}_D = 0 \quad (1 \leq i, j \leq 3). \quad (6)$$

The full finite-sum cancellation is recorded in the proof package and is independently expanded by the symbolic lane.

The two linear relations leave two generically independent integrals. A regular common fiber is closed in a compact energy shell. Liouville–Arnold therefore identifies each connected component with  $\mathbb{T}^2$  and gives  $\theta(t) = \theta(0) + t\omega$ . Return occurs exactly when

$$T\omega \in 2\pi\mathbb{Z}^2 \quad \text{for some } T > 0. \quad (7)$$

For two nonzero components this is  $\omega_1/\omega_2 \in \mathbb{Q}$ . Continuously selected resonant tori are not an isolated primitive-orbit ledger. Equations (4)–(7) are the *resolvent torus owner* of revision round one.

### 4 All declared degenerations and quantization

At  $x = \pm e_i, p = 0$ , a tangent component  $q_j, j \neq i$ , obeys

$$\ddot{q}_j = -(a_j - a_i)q_j.$$

Thus the two copies over  $e_1, e_2, e_3$  are respectively elliptic–elliptic, saddle–center, and saddle–saddle. Every face  $x_i = p_i = 0$  is invariant and reduces to

$$H_i = \frac{1}{2}\dot{\theta}^2 + \frac{1}{2}(a_j \cos^2 \theta + a_k \sin^2 \theta), \quad \{i, j, k\} = \{1, 2, 3\}.$$

If  $a_1 = a_2$ , the individual fractions  $F_1, F_2$  are not used. Equation (3) instead preserves the  $SO(2)$  Noether momentum  $J_{12} = L_{12}$ , while  $F_3$  remains nonsingular and  $SO(2)$  invariant. Thus, writing  $a_1 = a_2 = a \neq b = a_3$ ,

$$\{J_{12}, F_3\}_D = 0, \quad 2H = a + J_{12}^2 + (b - a)F_3. \quad (8)$$

This commuting pair is independent on a nonempty open set. Indeed, at  $x = (3/5, 0, 4/5)$  and  $p = (0, t, 0)$ , its differential determinant on the tangent variations  $((-4/5, 0, 3/5), 0)$  and  $(0, (0, 1, 0))$  is

$$-\frac{8(9(b-a) + 25t^2)}{125(b-a)},$$

which is nonzero after avoiding the sole possible exceptional value of  $t > 0$ . The other double faces follow by permutation. If  $A = aI$ , then  $\dot{p} = -|p|^2 x$ :  $p = 0$  is the equilibrium sphere and  $p \neq 0$  is a great circle of least period  $2\pi/|p|$ .

Fix a positive Planck parameter  $\hbar > 0$ . The canonical source quantization is

$$\widehat{H} = -\frac{\hbar^2}{2}\Delta_{S^2} + \frac{1}{2}x^T A x, \quad D(\widehat{H}) = H^2(S^2) \subset L^2(S^2). \quad (9)$$

The round Laplacian is self-adjoint with compact resolvent, and the real smooth potential is bounded. The bounded-perturbation theorem preserves both properties. The excluded value  $\hbar = 0$  would instead require the maximal  $L^2(S^2)$  domain for the bounded multiplication operator. No closed anisotropic spectrum is asserted.

## 5 Evidence, sources, and Route-A boundary

The exact ledger has 60 rational tangent-state rows, 120 rational resolvent probes, 30 axial rows, 30 coordinate-face rows, and explicit axisymmetric and isotropic boundaries. A producer-independent checker recomputes every fraction. A separate symbolic lane expands the Poisson brackets, resolvent identity, and Lax equation. These receipts test implementation conventions; the preceding proof handles the continuum.

The nearest workspace owners are C186 Euler top, C244 spherical pendulum, C313 round-sphere geodesics, and C344 resonant-triad elliptic dynamics. None has this holonomic anisotropic sphere potential and its Uhlenbeck–resolvent generator.

The conservative tuple is

$$(A0\_FAIL, A1\_WEAK, A2\_FAIL, A3\_FAIL, A4\_NATURAL\_QUANTIZATION).$$

There is no rational-prime carrier, logarithmic-prime clock, target Euler factor, root number, automorphy, target divisor or functional equation, target-zero match, or Hilbert–Polya operator. Route A is rejected and Route B is false. This is the *boundary and route firewall* of the final revision.

## Source boundary

Neumann owns the original mechanical problem; Moser owns the modern quadrics/spectral lineage; Knoerrer owns the geodesic correspondence cited here. This convention-locked reconstruction makes no priority claim.

## References

- [1] C. Neumann, “De problemate quodam mechanico, quod ad primam integralium ultraellipticorum classem revocatur,” *J. Reine Angew. Math.* 56 (1859), 46–63, DOI 10.1515/crll.1859.56.46.
- [2] J. Moser, “Geometry of Quadrics and Spectral Theory,” in *The Chern Symposium 1979*, Springer (1980), 147–188, DOI 10.1007/978-1-4613-8109-9\_7.
- [3] H. Knoerrer, “Geodesics on quadrics and a mechanical problem of C. Neumann,” *J. Reine Angew. Math.* 334 (1982), 69–78, DOI 10.1515/crll.1982.334.69.